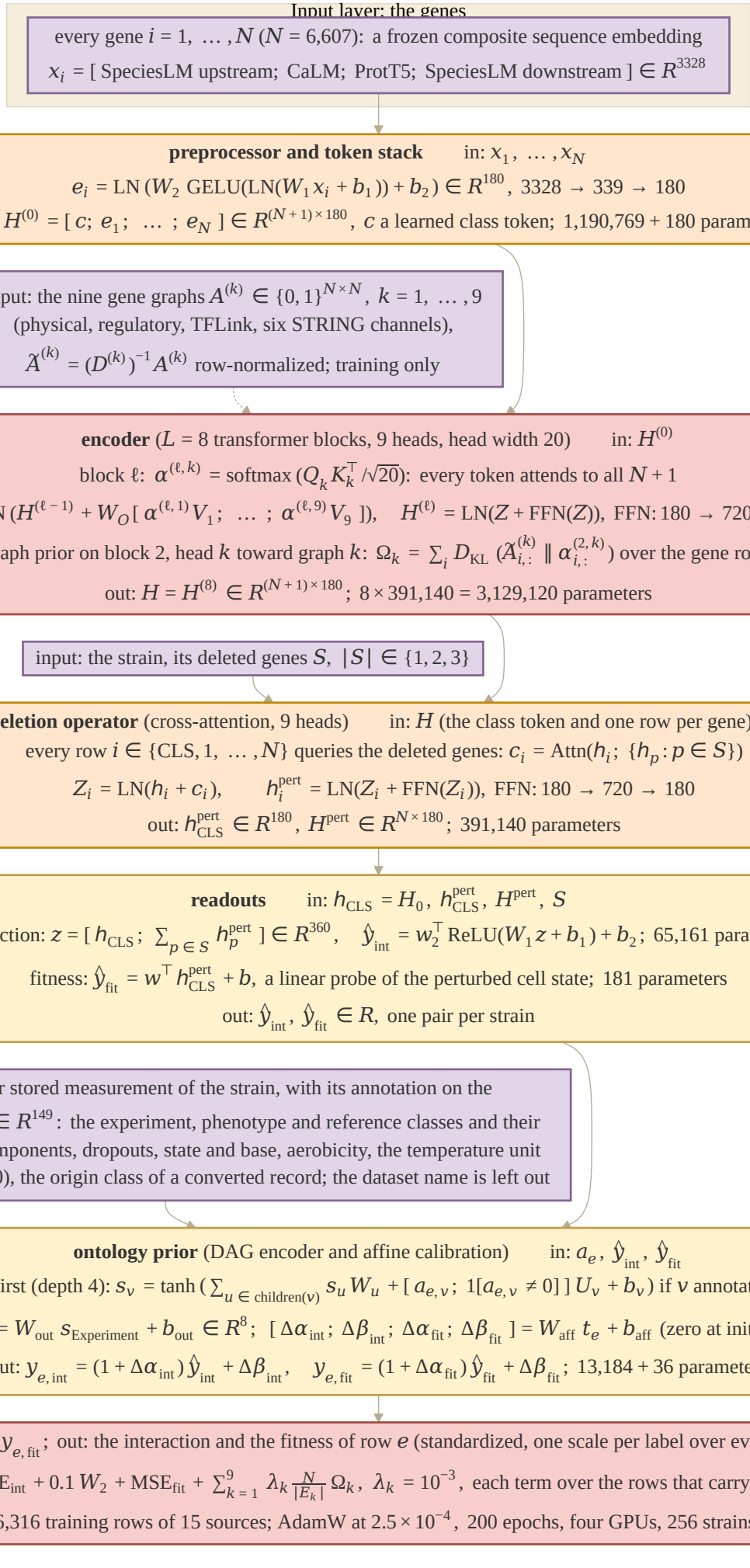

Input layer: the genes
every gene $i = 1, \ldots, N$ (N = 6,607): a frozen composite sequence embedding
x_i = [SpeciesLM upstream; CaLM; ProtT5; SpeciesLM downstream] $\in R^{3328}$
preprocessor and token stack in: $x_1, \ldots, x_N$
$e_i = \mathrm{LN}(W_2\ \mathrm{GELU}(\mathrm{LN}(W_1 x_i + b_1)) + b_2) \in R^{180}$, 3328 → 339 → 180
out: $H^{(0)} = [\, c; e_1; \ldots; e_N \,] \in R^{(N+1)\times 180}$, c a learned class token; 1,190,769 + 180 parameters
input: the nine gene graphs $A^{(k)} \in \{0,1\}^{N\times N}$, $k = 1, \ldots, 9$
(physical, regulatory, TFLink, six STRING channels),
$\tilde{A}^{(k)} = (D^{(k)})^{-1} A^{(k)}$ row-normalized; training only
encoder (L = 8 transformer blocks, 9 heads, head width 20) in: $H^{(0)}$
block ℓ: $\alpha^{(\ell,k)} = \mathrm{softmax}(Q_k K_k^\top / \sqrt{20})$: every token attends to all $N+1$
$Z = \mathrm{LN}(H^{(\ell-1)} + W_O[\, \alpha^{(\ell,1)} V_1; \ldots; \alpha^{(\ell,9)} V_9 \,])$, $H^{(\ell)} = \mathrm{LN}(Z + \mathrm{FFN}(Z))$, FFN: 180 → 720 → 180
graph prior on block 2, head k toward graph k: $\Omega_k = \sum_i D_{\mathrm{KL}}(\tilde{A}^{(k)}_{i,:} \,\|\, \alpha^{(2,k)}_{i,:})$ over the gene rows
out: $H = H^{(8)} \in R^{(N+1)\times 180}$; 8 × 391,140 = 3,129,120 parameters
input: the strain, its deleted genes S, $|S| \in \{1,2,3\}$
deletion operator (cross-attention, 9 heads) in: H (the class token and one row per gene), S
every row $i \in \{\mathrm{CLS}, 1, \ldots, N\}$ queries the deleted genes: $c_i = \mathrm{Attn}(h_i; \{h_p : p \in S\})$
$Z_i = \mathrm{LN}(h_i + c_i)$, $h_i^{\mathrm{pert}} = \mathrm{LN}(Z_i + \mathrm{FFN}(Z_i))$, FFN: 180 → 720 → 180
out: $h^{\mathrm{pert}}_{\mathrm{CLS}} \in R^{180}$, $H^{\mathrm{pert}} \in R^{N\times 180}$; 391,140 parameters
readouts in: $h_{\mathrm{CLS}} = H_0$, $h^{\mathrm{pert}}_{\mathrm{CLS}}$, H^{pert}, S
interaction: $z = [\, h_{\mathrm{CLS}}; \sum_{p\in S} h_p^{\mathrm{pert}} \,] \in R^{360}$, $\hat{y}_{\mathrm{int}} = w_2^\top \mathrm{ReLU}(W_1 z + b_1) + b_2$; 65,161 parameters
fitness: $\hat{y}_{\mathrm{fit}} = w^\top h^{\mathrm{pert}}_{\mathrm{CLS}} + b$, a linear probe of the perturbed cell state; 181 parameters
out: $\hat{y}_{\mathrm{int}}, \hat{y}_{\mathrm{fit}} \in R$, one pair per strain
input: one row e per stored measurement of the strain, with its annotation on the
experiment ontology $a_e \in R^{149}$: the experiment, phenotype and reference classes and their
parents, the medium's components, dropouts, state and base, aerobicity, the temperature unit
and value (celsius over 100), the origin class of a converted record; the dataset name is left out
ontology prior (DAG encoder and affine calibration) in: a_e, $\hat{y}_{\mathrm{int}}$, $\hat{y}_{\mathrm{fit}}$
every node v, leaves first (depth 4): $s_v = \tanh(\sum_{u\in \mathrm{children}(v)} s_u W_u + [\, a_{e,v};\ 1[a_{e,v} \neq 0] \,] U_v + b_v)$ if v annotated or above one, else 0
token $t_e = W_{\mathrm{out}} s_{\mathrm{Experiment}} + b_{\mathrm{out}} \in R^8$; $[\, \Delta\alpha_{\mathrm{int}}; \Delta\beta_{\mathrm{int}}; \Delta\alpha_{\mathrm{fit}}; \Delta\beta_{\mathrm{fit}} \,] = W_{\mathrm{aff}} t_e + b_{\mathrm{aff}}$ (zero at initialization)
out: $y_{e,\mathrm{int}} = (1 + \Delta\alpha_{\mathrm{int}})\hat{y}_{\mathrm{int}} + \Delta\beta_{\mathrm{int}}$, $y_{e,\mathrm{fit}} = (1 + \Delta\alpha_{\mathrm{fit}})\hat{y}_{\mathrm{fit}} + \Delta\beta_{\mathrm{fit}}$; 13,184 + 36 parameters
in: $y_{e,\mathrm{int}}$, $y_{e,\mathrm{fit}}$; out: the interaction and the fitness of row e (standardized, one scale per label over every source)
$L = \mathrm{MSE}_{\mathrm{int}} + 0.1\, W_2 + \mathrm{MSE}_{\mathrm{fit}} + \sum_{k=1}^{9} \lambda_k \frac{N}{|E_k|} \Omega_k$, $\lambda_k = 10^{-3}$, each term over the rows that carry the label
1,046,316 training rows of 15 sources; AdamW at 2.5×10^{-4}, 200 epochs, four GPUs, 256 strains each